Mock Paper A(2)

Started on	Thursday, 21 June 2018, 8:33 AM
State	Finished
Completed on	Thursday, 21 June 2018, 8:33 AM
Time taken	13 secs
Grade	**0.00** out of 100.00
Feedback	You are on the right track. Could you spend some more time on your revision?

Question 1
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 032
Phenothiazines that are associated with pigment deposits in anterior lens capsule include

Select one:
- Chlorpromazine and promazine
- perphenazine and trifluoperazine
- Chlorpromazine and perphenazine
- promazine and thioridazine
- Chlorpromazine and thioridazine

Your answer is incorrect.
Aliphatic phenothiazines chlorpromazine and thioridazine are associated with pigmentary retinopathy and lens pigmentation.
The correct answer is: Chlorpromazine and thioridazine

Question 2
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 049
Which of the following is a risk factor for neuroleptic-induced akathisia?

Select one:
- Non-affective psychosis
- Afro-Caribbean ethnicity
- Younger age
- lipid abnormalities
- Female gender

Your answer is incorrect.
Female gender and older age are risk factors for akathisia.
The correct answer is: Female gender

Question 3
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 058
Which of the following can provide an acute symptom relief for a patient with tardive dyskinesia?

Select one:
- IV amantadine
- Bromocriptine
- Increase neuroleptic dose
- IV amphetamine
- Stopping all neuroleptics

Your answer is incorrect.
Among the given options, increasing neuroleptic dose can provide acute relief but can worsen TD in the long run and so is not recommended. Stopping all neuroleptics suddenly can worsen TD acutely.
The correct answer is: Increase neuroleptic dose

Question 4
Not answered
Marked out of 1.00

MoK Adverse Effects 059
The most well-established risk factor in terms of developing late onset dyskinesia when using antipsychotics is

Select one:

Flag question

- brain pathology
- mood disorder
- increasing age
- Afro Caribbean ethnicity
- female gender

Your answer is incorrect.

One of the risk factors for TD is age. The majority of those with tardive dyskinesia are over the age of 55.

The correct answer is: increasing age

Question 5

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 063
Fetal Ebstein's anomaly is associated with which of the following psychotropic medication?

Select one:

- Lithium
- Sodium valproate
- Lamotrigine
- Carbamazepine
- Gabapentin

Your answer is incorrect.

Lithium use during pregnancy has a well-known association with the cardiac malformation Ebstein's anomaly (relative risk is 10-20 times more than control, but the absolute risk is low at 1:1000). The period of maximum risk to the foetus on exposure is 2-6 weeks after conception.

The correct answer is: Lithium

Question 6

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 065
A woman comes into Emergency Department with lethargy, nausea and diarrhoea with weight loss on a hot afternoon. She is taking some unknown psychotropic medication for a long time. She is recently started on 'water tablets' for swelling in her legs. Which of the following is a likely cause of her symptoms?

Select one:

- Hypokalemia
- Lithium toxicity
- Hyponatremia
- Neuroleptic malignant syndrome
- Serotonin syndrome

Your answer is incorrect.

The story here is suggestive of lithium overdose or toxicity.

The correct answer is: Lithium toxicity

Question 7

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 068
The factors that increase the risk of agranulocytosis in patients treated with clozapine include all of the following except

Select one:

- Female gender
- Male gender
- Increasing age
- Low baseline WBC count
- Asian ethnicity

Your answer is incorrect.

The factors that increase the risk for agranulocytosis in patients treated with clozapine are Jewish descent, female gender (male-to-female ratio of 1:2), increasing age, first 6 months of treatment, and a low white blood count at baseline.

The correct answer is: Male gender

Question 8

Not answered

MoK Adverse Effects 073
Which one of the following is the least likely feature associated with Neuroleptic malignant syndrome?

Marked out of 1.00

Flag question

Select one:

- Muscle rigidity
- Hyperthermia
- Autonomic lability
- myoclonus
- Raised CPK levels

Your answer is incorrect.

Myoclonus is a feature that could be seen in serotonin syndrome and not in NMS.

The correct answer is: myoclonus

Question 9

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 074
The most common side effect of sildenafil is

Select one:

- Bradycardia
- Hypotension
- Nasal stuffiness
- Insomnia
- Hair loss

Your answer is incorrect.

Common side effects of sildenafil citrate (occurring in more than 2 percent of people) includes headache 16%, facial flushing-10 %, indigestion- 7 %, nasal congestion- 4 %, Urinary tract Infections (UTI) -3 %, vision problems- 3%.

The correct answer is: Nasal stuffiness

Question 10

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 075
Tom recently underwent an alcohol detoxification programme. Upon achieving abstinence, he was keen on anti-craving treatment and has been started on Acamprosate. Which one among the following is a common side effect of Acamprosate?

Select one:

- Constipation
- Hair loss
- Nausea
- Headache
- Dizziness

Your answer is incorrect.

Acamprosate is well tolerated though side effects include diarrhoea, abdominal pain, nausea and vomiting, also pruritus. It should be initiated as soon as abstinence has been achieved and should be continued for up to 6 months, with regular monthly supervision. (Ref: The Maudsley Prescribing Guidelines, 11th ed. Pg. 370)

The correct answer is: Nausea

Question 11

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 076
Which one among the following has a high hepatotoxic potential?

Select one:

- Trazadone
- Amitryptyline
- Dothiepin
- Imipramine
- Nefazodone

Your answer is incorrect.

Because of concerns of hepatotoxicity, the sale of the antidepressant nefazodone hydrochloride was discontinued by the manufacturer in 2003.

The correct answer is: Nefazodone

Question 12

Not answered

MoK Adverse Effects 077
Low serum iron is associated with which of the following side effects related to neuroleptic medication treatment?

Marked out of 1.00

Flag question

Select one:

- sedation
- orthostatic hypotension
- weight gain
- neuroleptic malignant syndrome
- hyperglycaemia

Your answer is incorrect.

Low serum iron is associated with the neuroleptic malignant syndrome and extrapyramidal reactions. Correcting with supplemental iron may reduce the risk for these conditions.

The correct answer is: neuroleptic malignant syndrome

Question 13

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 078
A 45-year-old gentleman with depression has been started on Tranylcypromine. Which drug is safe to be taken?

Select one:

- Nifedipine
- Pethidine
- Pseudoephedrine
- Buprenorphine
- Salbutamol

Your answer is incorrect.

Tranylcypromine is a MAOI Inhibitor. MAOIs should not be combined with other psychoactive substances (antidepressants, painkillers, stimulants, both legal and illegal etc.). Certain combination of drugs are potentially fatal with MAOIs: examples include SSRIs, tricyclics, MDMA, meperidine, tramadol and dextromethorphan.

The correct answer is: Nifedipine

Question 14

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 079
According to CATIE study, which drug had the lowest drop out rate when treating schizophrenia?

Select one:

- Amisulpride
- Quetiapine
- Olanzapine
- Risperidone
- Ziprasidone

Your answer is incorrect.

CATIE stands for Clinical Antipsychotic Trials of Intervention Effectiveness. The study design was double-blind pragmatic RCT. 1493 patients with chronic schizophrenia were included and randomised to one of the following: Olanzapine, Quetiapine, Risperidone, Ziprasidone (added later in the trial), Perphenazine. The primary outcome was a 'real-world' measure - discontinuation for any reason (irrespective of whether patient-initiated or physician-initiated). Irrespective of the prescribed drug - 74% discontinued treatment in 18 months (surprisingly high despite naturalistic design). Median time to discontinue was 4.6 months. Olanzapine had lowest discontinuation rate (still 64%) - but highest side effect burden. 64% discontinued olanzapine; 75%, perphenazine; 82%, quetiapine; 74%, risperidone; and 79%, ziprasidone.

The correct answer is: Olanzapine

Question 15

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 080
Which of the following adverse effect of clozapine is mediated via alpha adrenoceptor blockade?

Select one:

- sedation
- weight gain
- constipation
- sialorrhea
- orthostatic hypotension

Your answer is incorrect.

Alpha 1 blockade leads to a postural drop in BP.

The correct answer is: orthostatic hypotension

Question 16
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 038
Mr Y is a 45-year-old gentleman with history of schizophrenia. He suffers from akathisia due to taking risperidone and also has some depressive symptoms. The drug of choice in his case would be

Select one:
- Venlafaxine
- Amitrytyline
- Paroxetine
- Fluoxetine
- Mirtazapine

Your answer is incorrect.

In the treatment of akathisia, a reduction in symptoms may be seen with mirtazapine. Other drugs, which are helpful, include propranolol (30-80 mg/day), low dose clonazepam, 5-HT2 antagonists such as cyproheptadine, mianserin, trazodone and diphenhydramine.

The correct answer is: Mirtazapine

Question 17
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 041
Buprenorphine acts on which receptor?

Select one:
- Sigma
- Mu
- Kappa
- Delta
- Alpha

Your answer is incorrect.

Buprenorphine is a μ-opioid receptor agonist with high affinity, but a low intrinsic activity. Compared with morphine which behaves as a full μ-opioid agonist, buprenorphine is usually defined as a partial μ-opioid agonist that shows high affinity for and slow dissociation from the μ-opioid receptor.

The correct answer is: Mu

Question 18
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 042
A 40-year-old woman with a past history of psychotic depression is having a relapse but is managed by the Crisis and Home Treatment team in the community. She is currently on lofepramine 180 mg and risperidone 2 mg at bedtime. After a week, she reports to the community worker who visited her at home that her nipples are leaking. She is very concerned and denies she could be pregnant. Which class of medications is most commonly known to cause this problem?

Select one:
- Antidepressants of SSRI class
- Tricyclic antidepressants
- Mood stabilisers
- Antidepressants of MAO inhibitors class
- Neuroleptics

Your answer is incorrect.

Dopamine receptor blockade causes hyperprolactinemia, which can cause breast enlargement. This is associated with galactorrhea in women and suppression of testosterone production in men. The typical neuroleptics such as haloperidol and chlorpromazine are particularly prone to cause an increase in prolactin.

The correct answer is: Neuroleptics

Question 19
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 043
A 50 year old man is treated for erectile dysfunction using sildenafil as and when required. Availability of which of the following vasodilatory substances is increased by sildenafil?

Select one:
- Nitric oxide
- Endothelin
- Phosphodiesterase
- Angiotensin
- Prostacyclin

Your answer is incorrect.

By inhibiting phosphodiesterase, sildenafil increases the availability of nitric oxide, a second messenger that acts as vasodilator of penile smooth muscle producing an erection.

The correct answer is: Nitric oxide

Question 20

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 044
Which of the following substrate/receptors blockade explains the clinical efficacy of riluzole in motor neuron disease?

Select one:

- Acetyl cholinesterase enzyme
- AMPA receptors
- NMDA receptors
- GABA receptors
- Dopamine D5 receptor

Your answer is incorrect.

Riluzole is an NMDA antagonist, similar to memantine

The correct answer is: NMDA receptors

Question 21

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 045
The mechanism of action of Memantine is

Select one:

- Non competitive NMDA receptor agonist
- Non competitive NMDA receptor antagonist
- Competitive NMDA receptor antagonist
- Competitive NMDA receptor agonist
- Acetylcholinesterase inhibitor

Your answer is incorrect.

The mechanism of memantine involves blockade of N-methyl-d-aspartate (NMDA) glutamate receptors. Unlike ketamine which is a high-affinity noncompetitive blocker, memantine is a non-competitive blocker with low affinity and binds only to actively open NMDA channels. Its receptor dissociation rate is relatively fast, and so it does not accumulate and interfere with normal NMDA activity

The correct answer is: Non competitive NMDA receptor antagonist

Question 22

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 046
What is the mechanism of action of atomoxetine when used in the treatment of ADHD?

Select one:

- Dopaminergic agonism
- Noradrenaline reuptake inhibition
- Glutamatergic antagonism
- Serotonin reuptake inhibition
- Dopamine reuptake inhibition

Your answer is incorrect.

The exact mechanism of action is unknown, but atomoxetine selectively inhibits the pre-synaptic noradrenaline transporter (NARI - Noradrenaline reuptake inhibitor)

The correct answer is: Noradrenaline reuptake inhibition

Question 23

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 047
Which one among the following properties is not a significant effect of Trazodone?

Select one:

- Serotonin reuptake blockade
- Antihistamine properties
- Strong alpha-1 antagonism
- Norepinephrine reuptake blockade
- Serotonin 2 antagonism

Your answer is incorrect.

Nefazadone achieves its clinical effects by inhibiting the reuptake of primarily serotonin but also norepinephrine and by antagonizing the postsynaptic 5-HT2 receptor and a1-adrenergic receptors. Trazadone works via serotonin 2 antagonism and serotonin re-uptake blockade with strong alpha-1 antagonism and antihistamine properties.

The correct answer is: Norepinephrine reuptake blockade

Question 24
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 047
The effect of which of the following groups of drugs is least influenced by their pharmacokinetics?

Select one:
- SSRIs
- MAO Inhibitors
- Benzodiazepines
- Neuroleptics
- Tricyclics

Your answer is incorrect.

MAOIs are also well absorbed from the gastrointestinal tract. Their short half-life of one to two hours is not particularly relevant as they bind irreversibly with MAO. Thus, the activity of these drugs depends less on pharmacokinetics and more on the synthesis of new MAO to restore normal enzyme activity. This synthesis requires approximately two weeks. This class of drugs is little used due to their potentially dangerous interactions with sympathomimetics and foods containing tyramine.

The correct answer is: MAO Inhibitors

Question 25
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 048
Which of the following psychotropic drugs could be potentially affected by acetylator status?

Select one:
- Phenothiazines
- Lithium
- Tricyclic antidepressants
- Acetylcholinesterase inhibitor
- MAO inhibitors

Your answer is incorrect.

Though potentially MAO inhibitors can be affected acetylator status, in practical use this is not usually a problem.

The correct answer is: MAO inhibitors

Question 26
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 049
Which one among the following antidementia drugs is mainly renally excreted?

Select one:
- Tacrine
- Donepezil
- Rivastigmine
- Galantamine
- Memantine

Your answer is incorrect.

Memantine is mainly renally excreted. But as robust data from renal impaired patients are not available, drug dose should be reduced in patients with mild to moderate renal impairment. Memantine is not recommended for patients with severe renal impairment (Jann & Small 2005).

The correct answer is: Memantine

Question 27
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 050
Which drug acts as a full agonist at Mu receptor and has a long half life?

Select one:
- Buspirone
- Buprenorphine
- Disulfiram
- Methadone

- Acamprosate

Your answer is incorrect.

Methadone is a full mu- opioid agonist. Its half-life is approximately 24 hours with regular use.

The correct answer is: Methadone

Question 28

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 051
When swapping MAOIs to SSRIs, what is the optimum wash out period?

Select one:

- Minimum 2 days is essential
- Minimum 5 days is essential
- Minimum 2 months is essential
- Minimum 2 weeks is essential
- Minimum 5 weeks is essential

Your answer is incorrect.

MAOI to SSRI- 2 weeks SSRI to MAOI-2 weeks for all, 5 weeks for Fluoxetine TCA to MAOI- 2 weeks MAOI to TCA- 1-2 weeks.

The correct answer is: Minimum 2 weeks is essential

Question 29

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 052
Apparent volume of distribution of haloperidol is

Select one:

- 1l/kg
- 0.5l/kg
- 0.1l/kg
- 0.25l/kg
- more than 2l/kg

Your answer is incorrect.

The apparent volume of distribution is the theoretical volume of fluid into which the total drug administered would have to be diluted to produce the concentration in plasma. For example, if 1000 mg of a drug is given and the subsequent plasma concentration is 10 mg/L, that 1000 mg seems to be distributed in 100 L (dose/volume = concentration; 1000 mg/x L = 10 mg/L; therefore, x = 1000 mg/10 mg/L = 100 L). Many acidic drugs (e.g., warfarin) are highly protein-bound and thus have a small apparent volume of distribution. Many basic drugs (e.g., amphetamine) are extensively taken up by tissues and thus have an apparent volume of distribution larger than the volume of the entire body. Haloperidol is very extensively bound to plasma proteins (90%). It has a volume of distribution ranging from 9.5 to 21.7 liters/kilogram (Kudo and Ishizaki, 1999) [Drug Distribution to Tissues - Pharmacokinetics - Merck. Retrieved from http://www.merckmanuals.com/professional/clinical-pharmacology/pharmacokinetics/].

The correct answer is: more than 2l/kg

Question 30

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 053
Among Caucasians who are social drinkers, what percentage of alcohol is metabolized by alcohol dehydrogenase?

Select one:

- less than 9
- 60-79
- more than 90%
- 75-85
- 40-59

Your answer is incorrect.

The major site of alcohol metabolism or breakdown (biotransformation) is in the liver. Alcohol dehydrogenase is the rate-limiting enzyme in the metabolism of alcohol and thus limits the amount of alcohol metabolized. More than 90% of consumed alcohol in Caucasians is metabolized via this route. Another significant metabolic pathway in the liver is called microsomal ethanol oxidizing system (MEOS). This consists of the induction of the liver enzyme cytochrome P-450 (CYP2E1) which biotransforms a number of substances including alcohol and barbiturates

The correct answer is: more than 90%

Question 31

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 054
Which of the following over-the-counter preparations must be avoided by a patient who is prescribed phenelzine for resistant depression with social anxiety?

Select one:

- Nasal decongestant sprays
- Artificial tears
- Antacids
- Nicotine inhalers
- Heat patches

Your answer is incorrect.

If MAOIs are taken with decongestants, the patient may experience a life-threatening rise in blood pressure due to synergistic sympathetic drive.

The correct answer is: Nasal decongestant sprays

Question 32

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 056

Which of the following statements about lithium is correct?

Select one:

- Lithium levels fall within 10 days of starting a thiazide diuretic
- NSAIDs can increase the plasma levels of lithium to >40%
- A third of patients treated with lithium experience at least one side effect
- 70% of middle aged women develop hypothyroidism after 6-12 months of treatment
- ACE inhibitors cause a reduction in lithium levels

Your answer is incorrect.

NSAIDs increase the plasma levels of lithium (>10 to >40% in some cases) by inhibiting the synthesis of renal prostaglandins and reducing renal blood flow. This increases renal reabsorption of both sodium and lithium. The levels of lithium usually increase within ten days of a thiazide diuretic being prescribed. ACE inhibitors can reduce thirst, which can lead to dehydration, and increase renal sodium loss, leading to increased sodium reabsorption by the kidneys, causing an increase in lithium plasma levels. About 20% of middle-aged women develop hypothyroidism after 6-12 months of treatment. Approximately two-thirds of patients treated with lithium experience at least one side effect.

The correct answer is: NSAIDs can increase the plasma levels of lithium to >40%

Question 33

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 042

Which of the following subtests of Wechsler Adult Intelligence Scale (WAIS) is noted to resist any decline with aging?

Select one:

- Digit symbol
- Block design
- Digit span
- similiarities
- Picture completion

Your answer is incorrect.

Hold tests' are supposed to be resistant to age-related decline; so they may be sensitive to organic brain damage such as dementia

The correct answer is: Picture completion

Question 34

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 057

Hayling test in neuropsychological assessments is used to test which of the following?

Select one:

- Digit span and executive function
- Verbal fluency and performace IQ
- Verbal fluency and executive function
- Premorbid IQ
- Verbal memory

Your answer is incorrect.

The Hayling test (Burgess and Shallice, 1997) is an instrument that asks the subject to complete sentences by saying a final word to complete the sentence. The aim of Part A of the test is evaluate the preprogrammed organizing process because in this phase the constructs to be examined are concentrated attention, verbal initiation, processing speed, and the strategy of a well-succeeded search for automated words related to the preactivation of semantic networks. Part B assesses more complex Executive Function components, such as verbal inhibition and planning, because the subject must inhibit the content of the sentence (semantics) and develop alternative strategies in his lexical search to complete the syntactic stimulus (Chan et al., 2008) (Retrieved from http://www.psycneuro.org/index.php/path/article/viewFile/104/393).

The correct answer is: Verbal fluency and executive function

Question 35
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 058
A 49-year-old patient with schizophrenia scored poorly on Brixton test. This shows deficits in

Select one:
- Attention and concentration
- Verbal memory
- Visuospatial sequencing
- Abstraction
- Language

Your answer is incorrect.

The Brixton test is a visuospatial sequencing task that measures the ability to learn rules from sequences of stimuli. It is relatively free from educational and sociocultural bias as it does not require verbal responses. It can be administered in five and ten minutes. It is often used to assess executive functioning in people with neurological disorders such as strokes, dementia and psychiatric disorders such as schizophrenia.

The correct answer is: Visuospatial sequencing

Question 36
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 059
During extended cognitive assessment by a neuropsychologist, the subject is asked to read a short story containing 25 elements and the psychologist tested both immediate and delayed recall after 30 minutes. The test done by the neuropsychologist is called

Select one:
- Halstead - Reitan scale
- Rivermead behavioural memory test
- Wechsler memory scale
- Wechsler adult intelligence scale
- Rey-Osterrieth test

Your answer is incorrect.

Here the subject is asked to read a short story from the Wechsler Memory scale containing 25 elements and both immediate and delayed recall after an interval of 30 minutes is tested. This is one of the verbal subtests used in Wechsler memory scale. WMS-3 consists of tests grouped under two domains; verbal and nonverbal, all of which have immediate and delayed recall and or recognition components. It also has a third group of tests involve .mental control subtests'.

The correct answer is: Wechsler memory scale

Question 37
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 060
A psychologist is performing neuropsychological testing in a 45 year old man. Which one of the following is a test of set shifting?

Select one:
- Wechsler memory scale
- Ravens progressive matrices
- Boston naming test
- Wisconsin Card sorting test
- Clock drawing task

Your answer is incorrect.

Wisconsin Card Sorting test is a neuropsychological test of 'set shifting'. Shifting set refers to the process of updating or "shifting" cognitive strategies in response to changes in the environment. For example, in WCST patients are first asked to perform according to some rule (sort cords by number). Next, patients are asked to switch to a new rule (sort cords by color). Successful performance requires the ability to abandon an old strategy and start responding according to a new rule. Patients with frontal lobe damage are often impaired at tasks, which require set-shifting abilities.

The correct answer is: Wisconsin Card sorting test

Question 38
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 061
A 50-year-old man on a neurology ward has non-dominant parietal lobe dysfunction. Identify the test that he is least likely to be able to do?

Select one:
- Reading a sentence
- Serial subtraction of 7s from 100
- Copying intersecting pentagons
- Writing a sentence
- Correctly stating current location

Your answer is incorrect.

The cognitive process tested by asking the patient to copy intersecting pentagons is constructional praxis, which is a feature of non-dominant parietal lobe lesions. The effects of lesion involving non-dominant parietal lobe include visuospatial dysfunction, topographical memory loss, and constructional / dressing apraxia. These disorders could be seen in lesions of either hemisphere but observed more frequently and are of greater severity with lesions of the non-dominant parietal lobe.

The correct answer is: Copying intersecting pentagons

Question 39

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 062
Mr. Smith was seen by a clinical psychologist at a memory clinic and was noted to have features of executive dysfunction. Executive function is tested using which of the following tests?

Select one:

- National Adult Reading Test
- Raven's Progressive Matrices
- Wisconsin Card Sorting Test
- Vocabulary Test
- Rey-Osterrieth Test

Your answer is incorrect.

Executive function includes planning, initiation, sequencing, coordinating, error detection, error correction, set shifting, and termination. It is closely allied to other frontal functions such as judgement, problem-solving, impulse control, and abstract reasoning. These can be tested formally using Wisconsin Card Sorting Test.

The correct answer is: Wisconsin Card Sorting Test

Question 40

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 063
Non-fluent expressive dysphasia is seen in which of the following conditions

Select one:

- Frontotemporal dementia
- Lewy body dementia
- Progressive supranuclear palsy
- Parkinson's disease
- Cruetzfeldt Jacob disease

Your answer is incorrect.

Broca's aphasia is also called non-fluent expressive dysphasia. There is reduced frequency and fluency of speech and presents with few disjointed words. Comprehension is relatively preserved. It occurs with damage to Broca's area in the left frontal lobe. It is commonly seen in frontotemporal dementia (Pick's disease).

The correct answer is: Frontotemporal dementia

Question 41

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 064
Digit span test is administered to patient with suspected cognitive impairment. This test detects problem in which of the following cognitive faculties?

Select one:

- Semantic memory
- Procedural memory
- Premorbid IQ
- Working memory
- Calculation ability

Your answer is incorrect.

Digit span is a test of working memory (immediate), and attention

The correct answer is: Working memory

Question 42

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 049
Which of the following can be termed as an intermediate phenotype with high occurrence in relatives but cannot be classified as a subclinical endophenotype for schizophrenia?

Select one:

- Impaired attentional performance

- Working memory deficits
- Eye movement dysfunctions
- Impaired p50 suppression
- Neurological soft signs

Your answer is incorrect.

Neurological soft signs are observable intermediate phenotypes unlike the rest of the listed which are endophenotypes.

The correct answer is: Neurological soft signs

Question 43

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 055
What is the best test for diagnosing Huntington's disease?

Select one:

- CT scan
- Serum Protein Assay
- Polymerase Chain Reaction
- Karyotyping
- MRI scan

Your answer is incorrect.

Serum polymerase chain reaction (PCR) is the test of choice to examine the number of trinucleotide repeats (More than35 in adults and More than50 in children) in order to diagnose Huntington's disease (HD). The causative gene resides on the short arm of chromosome 4 at 4p16.3. A chromosomal karyotype can reveal only macroscopic defects in chromosomes such as deletions, translocations, or trisomies.

The correct answer is: Polymerase Chain Reaction

Question 44

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 061
The form of Down's syndrome, which is as likely to have come from father as that of mother's chromosome, is;

Select one:

- Mosaicism
- Duplication
- Robertsonian translocation
- Deletion
- Full trisomy 21

Your answer is incorrect.

In a Robertsonian translocation, two acrocentric chromosomes break at their centromeres. The resulting long arms fuse to form a single chromosome with a single centromere. The short arms also join to form a reciprocal product, which typically contains nonessential genes and is usually lost after a few cell divisions. When chromosome 21 is involved in a Robertsonian translocation (often with chromosome 14), this may result in Down's syndrome, which is as likely to have come from the father as that of the mother's chromosome.

The correct answer is: Robertsonian translocation

Question 45

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 062
Where does translation take place in human cells?

Select one:

- Nucleus
- Mitochondria
- Smooth endoplasmic reticulum
- Rough endoplasmic reticulum
- Golgi complex

Your answer is incorrect.

This answer is not precise, but best of the rest. In fact it takes place in ribosomes - Rough ER harbours these ribosomes on its outer surface when translation takes place.

The correct answer is: Rough endoplasmic reticulum

Question 46

Not answered

MoK Molecular Genetics 063
Smooth pursuit eye tracking dysfunction is proposed as a strong endophenotype marker for which of the following group of disorders?

Marked out of 1.00

Flag question

Select one:

- Personality disorders
- OCD spectrum disorders
- Mania
- Schizophrenia
- Anxiety disorders

Your answer is incorrect.

Several studies have shown that eye tracking dysfunction is present in a high percentage of subjects with schizophrenia and their non-schizophrenic relatives http://ajp.psychiatryonline.org/cgi/content/full/162/3/466

The correct answer is: Schizophrenia

Question 47

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 064
Which of the following personality disorders has the strongest support for a genetic contribution?

Select one:

- Dependent perosnality
- Schizoid disorder
- Anankastic disorder
- Antisocial disorder
- Schizotypal disorder

Your answer is incorrect.

Though schizotypal, antisocial, dependent and avoidant the with familial risk, molecular and twin study evidences so far strongly implicate genetic aetiology for the schizotypal disorder.

The correct answer is: Schizotypal disorder

Question 48

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 065
A child with a deficiency of Hypoxanthine-guanine phosphoribosyltransferase developed increased acid. Which of the following behavior is consistent with this syndrome?

Select one:

- Self Injury
- Stereotypes
- Tangential speech
- Visual neglect
- Isolated performance IQ defect

Your answer is incorrect.

Lesch-Nyhan syndrome (LNS), is a rare disorder caused by a deficiency of the enzyme hypoxanthine-guanine phosphoribosyl transferase (HGPRT). Persons affected are cognitively impaired; the condition is characterized by the overproduction and accumulation of uric acid, a waste product of normal chemical processes that is found in blood and urine. The uncontrollable self-injury a usually begins at three years of age. The self-injury begins with biting of the lips and tongue.

The correct answer is: Self Injury

Question 49

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 066
Mutations that occur in body cells may not get transmitted via gametes. These are called

Select one:

- polymorphisms
- missense mutations
- Somatic mutations
- pseudomutations
- Nonsense mutations

Your answer is incorrect.

Mutations that occur in somatic cells are somatic mutations and unlike germ cell mutations these may not be transmitted to next generation.

The correct answer is: Somatic mutations

Question 50
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 067
Trinucleotide repeat expansions in DNA can be detected using

Select one:
- Northern blot
- ELISA
- Polymerase chain reaction
- Western blot
- Southern blot

Your answer is incorrect.

As trinucleotide repeats are DNA abnormalities, the Southern blot can be used to detect them.

The correct answer is: Southern blot

Question 51
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 068
ApoE polymorphisms are associated with risk of Alzheimer's disease. Apo E4 alleles are also implicated in following disorders except

Select one:
- Sporadic CJD
- Parkinson's disease
- Intracerebral haemorrhage
- Recovery from head injury
- Herpes simplex encephalitis

Your answer is incorrect.

Associations between the E4 allele and Alzheimer disease, Parkinson disease, susceptibility to herpes simplex virus infection, poor recovery from head injury, intracerebral haemorrhage, and elective cardiac bypass surgery have been shown (Wright et al. Neurogenetics II: complex disorders, JNNP; 76 (5): 623, http://jnnp.bmj.com/content/76/5/623.full).

The correct answer is: Sporadic CJD

Question 52
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 069
Which of the following statements about epidemiology of Autism is true?

Select one:
- It is more common in children who had MMR vaccine
- The heritability of this disorder is greater than 90%
- The monozygotic twin concordance is more than 95%
- It is seen among children of parents from high socioeconomic status
- It is more common in girls than boys

Your answer is incorrect.

Twin studies have reported 60% concordance for classic autism in monozygotic twins versus 0 in dizygotic twins. If a broader autistic phenotype (that included communication and social disorders) is considered, the concordance increases remarkably from 60% to 92% in MZ twins and from 0 to 10% in DZ twins. This translates to 90% heritability.

The correct answer is: The heritability of this disorder is greater than 90%

Question 53
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 025
Which of the following cells can be described as macrophages of the central nervous system?

Select one:
- Microglia
- Oligodendrocytes
- Amacrine cells
- Endothelium
- Astrocytes

Your answer is incorrect.

Microglia are the macrophages of the brain; they are non-neuronal cells derived from mesoderm and help in scavenging the CNS.

The correct answer is: Microglia

Question 54
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 026
Which of the following arteries supplies Broca's area?

Select one:
- Posterior cerebral artery
- Anterior cerebral artery
- Vertebral artery
- Middle cerebral artery
- Lenticulostriate artery

Your answer is incorrect.

Broca's area lies in anterior aspect supplied by MCA. Wernicke's area is also supplied by the middle cerebral artery.
http://www.csuchico.edu/~pmccaffrey//syllabi/CMSD%20320/362unit11.html

The correct answer is: Middle cerebral artery

Question 55
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 028
The part of the brain involved in reward processing is

Select one:
- Amygdala
- Brain stem
- Nucleus accumbens
- Prefrontal cortex
- Hippocampus

Your answer is incorrect.

The amygdala - an almond-shaped group of nuclei at the heart of the telencephalon - has been associated with a range of cognitive functions, including emotion, learning, memory, attention and perception. Most current views of amygdala function emphasize its role in negative emotions, such as fear, and in linking negative emotions with other aspects of cognition, such as learning and memory. However, recent evidence supports a role for the amygdala in processing positive emotions as well as negative ones, including learning about the beneficial biological value of stimuli. Indeed, the amygdala's role in stimulus-reward learning might be just as important as its role in processing negative affect and fear conditioning (Baxter and Murray., The amygdala and reward. Nature Reviews Neuroscience. (2002). Retrieved from http://www.nature.com/nrn/journal/v3/n7/abs/nrn875.html). Comparative studies have implicated the nucleus accumbens (NAcc) in the anticipation of incentives, but the relative responsiveness of this neural substrate during anticipation of rewards versus punishments remains unclear. At the highest reward level, NAcc activation was correlated with individual differences in self-reported happiness elicited by the reward cues. These findings suggest that whereas other striatal areas may code for expected incentive magnitude, a region in the NAcc codes for expected positive incentive value (Excerpt from Knutson et al. J Neurosci. 2001 Aug 15;21(16):RC159. Retrieved from http://www.ncbi.nlm.nih.gov/pubmed/11459880).

The correct answer is: Nucleus accumbens

Question 56
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 029
Which of the following is the only cranial nerve without thalamic relay?

Select one:
- Trigeminal
- Optic
- Abducens
- Oculomotor
- Olfactory

Your answer is incorrect.

The olfactory nerve does not pass through the thalamus, unlike other cranial nerves.

The correct answer is: Olfactory

Question 57
Not answered
Marked out of 1.00
Flag question

MoK Neurochemistry 026
The rate limiting step in the synthesis of serotonin is mediated by which of the following enzymes?

Select one:
- L-tryptophan hydroxylase
- Monoamine oxidase A
- Tyrosine Hydroxylase
- Aminoacid decarboxylase
- Acetyl CO-A Synthetase

Your answer is incorrect.

The biochemical pathway for serotonin synthesis initially involves the conversion of L-tryptophan to 5-hydroxytryptophan by the enzyme L-tryptophan hydroxylase. This enzyme provides the rate limiting step in serotonin synthesis, in the same manner that norepinephrine and dopamine synthesis in adrenergic and dopaminergic neurons is controlled by the ability of the related enzyme, L-tyrosine hydroxylase, to convert L-tyrosine to L-dihydroxyphenyl-alanine (L-DOPA).

The correct answer is: L-tryptophan hydroxylase

Question 58
Not answered
Marked out of 1.00
Flag question

MoK Neurochemistry 028
In schizophrenia, dopamine deficiency is seen in which of the following brain regions?

Select one:
- All of the above
- Mesocortical projections
- Tuberoinfundibular pathway
- Nigrostriatum
- Mesolimbic pathway

Your answer is incorrect.

Excess dopamine is postulated as a cause of positive symptoms in the mesolimbic pathway.

The correct answer is: Mesocortical projections

Question 59
Not answered
Marked out of 1.00
Flag question

MoK Neurochemistry 029
Which of the following is an agonist at the mu-opioid receptor?

Select one:
- Disulfiram
- Naltrexone
- Buprenorphine
- Methadone
- Naloxone

Your answer is incorrect.

Buprenorphine is a partial opioid agonist.

The correct answer is: Methadone

Question 60
Not answered
Marked out of 1.00
Flag question

MoK Neurochemistry 030
The enzyme involved in degradation of noradrenaline in presynaptic neurons is

Select one:
- GABA transaminase
- Monoamine oxidase
- Homovanillic acid
- Vanillyl mandelic acid
- 3-methoxy, 4-hydroxy, phenylglycol

Your answer is incorrect.

The breakdown enzymes involved in degradation of noradrenaline in presynaptic neurons are monoamine oxidase (MAO-A) and catechol-o-methyl-transferase (COMT)

The correct answer is: Monoamine oxidase

Question 61
Not answered
Marked out of 1.00
Flag question

MoK Neurochemistry 031
Which of the following receptors requires the simultaneous binding of two different agonists for activation?

Select one:
- GABA B
- Muscarinic M3
- GABA A
- Mu Opiate
- NMDA

Your answer is incorrect.

NMDA receptors require simultaneous binding of two different agonists for their activity. NMDA ion channels open in response to the presence of both glutamate and glycine (co-activation molecules).

The correct answer is: NMDA

Question 62

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 015
Which of the following pathological bodies is made up of ubiquitin and alpha synuclein and is associated with loss of dopaminergic neurons in basal ganglia?

Select one:

- Hirano bodies
- Lewy bodies
- Neurofibrillary tangles
- Barr bodies
- Prion bodies

Your answer is incorrect.

Lewy bodies are seen Parkinsonian disorders and they are associated with a reduction of dopaminergic neurons (KC 1228)

The correct answer is: Lewy bodies

Question 63

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 019
A 55 year old lady presents with weakness of her legs and hands. She feels clumsy with a history of falls. On examination she has fasiculations in her tongue and memory deficits. She is diagnosed to have amyotrophic lateral sclerosis. The primary pathological change will be noted in

Select one:

- Motor neurons
- Neuromuscular junction
- Myocytes
- Spinal autonomic ganglion
- Sensory neurons

Your answer is incorrect.

ALS is a motor neuron disease where upper motor neurons are specifically damaged leading o loss of cells in anterior horns of cord and brain stem nuclei.

The correct answer is: Motor neurons

Question 64

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 020
Which of the following accounts for spongiform appearance in CJD?

Select one:

- Accumulation of prion proteins
- Neuronal vacuolation
- Trapped air in parenchymal cells
- Neuronal loss
- Neurophil loss

Your answer is incorrect.

Microscopically, CJD shows a shows a spongiform encephalopathy, secondary to neutrophil vacuolisation.

The correct answer is: Neuronal vacuolation

Question 65

Not answered

Marked out of 1.00

Flag question

MoK Neurophysiology 016
Miss. X suffers from eating disorder characterized by binge eating, body image disturbances and measures taken to reduce weight . This condition is associated with reduced levels of

Select one:

- Dopamine
- Glycine
- Vasopressin
- Leptin

- Endorphins

Your answer is incorrect.

Miss X is exhibiting symptoms of bulimia nervosa. Patients with bulimia nervosa as well as individuals who had recovered from bulimia nervosa had significantly lower serum leptin levels than healthy controls matched for BMI and percent body fat. The results are consistent with the hypothesis that decreased leptin function contributes to impaired postingestion satiety, neuroendocrine abnormalities, and abnormally low resting metabolic rate in bulimia nervosa. (Ref: (J Clin Endocrinol Metab 85: 4511-4514, 2000) and http://www.primarypsychiatry.com/aspx/articledetail.aspx?articleid=3532)

The correct answer is: Leptin

Question 66

Not answered

Marked out of 1.00

Flag question

MoK Neurophysiology 017
REM sleep disorder is associated with

Select one:

- Alzheimer's disease
- Fronto temporal dementia
- Wilson's disease
- Lewy body disease
- Huntington's disease

Your answer is incorrect.

REM sleep disorder is associated with disorders such as Parkinson's disease, multi-system atrophy and Gullian Barre syndrome. RBD may be the prodrome of neurodegenerative disorders such as DLB and Parkinson's disease.

The correct answer is: Lewy body disease

Question 67

Not answered

Marked out of 1.00

Flag question

MoK Neurophysiology 018
A 46-year-old lady suffers from excessive daytime somnolence and acute episodes of collapse on laughing. What is the most likely risk factor?

Select one:

- Binge drinking
- Low hypocretin (orexin) levels
- SSRI drug use
- Schizophrenia
- Gamma hydroxy butyric acid abuse

Your answer is incorrect.

This is a case of narcolepsy, where low orexin levels can be one of the common risk factors. Low CSF hypocretin-1 is highly specific (99.1%) and sensitive (88.5%) for narcolepsy with cataplexy

The correct answer is: Low hypocretin (orexin) levels

Question 68

Not answered

Marked out of 3.00

Flag question

MoK Pharmacology EMI018
Adverse effects
For each of the following drugs, identify the most important side effect associated with its therapeutic use

Galantamine: Choose...

Venlafaxine: Choose...

Topiramate: Choose...

Your answer is incorrect.

Antidementia drugs like galantamine can produce bradycardia, especially in those with supraventricular conduction delay.

A significant number of patients receiving doses above 300mg/day of venlafaxine experience an increase in diastolic blood pressure. This risk is not restricted to those with preexisting hypertension.

Topiramate is a weak inhibitor of renal carbonic anhydrase and can promote the development of renal stones. It is weight neutral and can even cause weight loss.

The correct answer is: Galantamine – Bradycardia, Venlafaxine – Diastolic Hypertension, Topiramate – Weight loss

Question 69

Not answered

Marked out of 3.00

MoK Pharmacology EMI020
Adverse effects
For each of the adverse effects given below, identify the drug most likely associated with it:

Flag question

Renal failure Choose...

Seizures Choose...

Nocturnal myoclonus Choose...

Your answer is incorrect.

Lithium: Renal damage may occur in severe, prolonged toxicity - but cumulative lithium use leads to renal failure more commonly. It takes at least ten years to develop interstitial fibrosis - the pathology behind the chronic renal damage.

SSRIs: Severe sweating especially nocturnally is seen in some patients; Terazosin is effective in counteracting sweating. Nocturnal myoclonus is reported with SSRIs. The repetitive leg movements are characterised by extensions of the large toe and flexion of all major leg joints (ankle, knee, and hips) and occur every 20 to 60 seconds. Benzodiazepines and levodopa may be tried.

Bupropion is associated with seizures and exacerbation of eating disorders due its anorexigenic properties.

The correct answer is: Renal failure – Lithium, Seizures – Bupropion, Nocturnal myoclonus – SSRIs

Question 70

Not answered

Marked out of 3.00

Flag question

HiY Neurosciences EMI019

Cell biology

Using descriptions given below, identify the type of enzymes from the list

Used in experiments to decrease gene expression Choose...

Provides a bridge between m-RNA and amino acids Choose...

Enzymes which attaches amino acids to t-RNA Choose...

Your answer is incorrect.

Explanation:

Small interfering RNA (Si-RNA) is a class of double-stranded RNA molecule, involving in RNA interference pathway, where it interferes with expression of a specific gene.

T-RNA is a small RNA molecule that transfers a specific active aminoacid to growing polypeptide chain at the ribosomal site of protein synthesis during translation. It also has a terminal site for amino acid attachment.

Aminoacyl t-RNA synthetase is an enzyme that catalyze the esterification of a specific amino acid or its precursors to one of all its compatible cognate t-RNA to form an aminoacyl t-RNA.

The correct answer is: Used in experiments to decrease gene expression – Si- RNA, Provides a bridge between m-RNA and amino acids – t-RNA, Enzymes which attaches amino acids to t-RNA – Aminoacyl t-RNA synthetase

Question 71

Not answered

Marked out of 3.00

Flag question

MoK Pharmacology EMI021

Selectivity of Tricyclics (TCAs)

Identify the most suitable drug, for the descriptions given below:

This drug is the most serotonergic reuptake selective of all tricyclic antidepressants Choose...

This drug is the most noradrenergic selective of all tricyclic antidepressants Choose...

This drug has the most antihistaminic activity. Choose...

Your answer is incorrect.

Clomipramine more serotonergic reuptake selective; desipramine is the most noradrenergic selective of TCAs. Amoxapine, nortriptyline, desipramine, and maprotiline (secondary amines) have the least anticholinergic activity; doxepin has the most antihistaminic activity.

The correct answer is: This drug is the most serotonergic reuptake selective of all tricyclic antidepressants – Clomipramine, This drug is the most noradrenergic selective of all tricyclic antidepressants – Desipramine, This drug has the most antihistaminic activity. – Doxepin

Question 72

Not answered

Marked out of 3.00

Flag question

MoK Pharmacology EMI022

Receptor mechanisms (1)

Using the definitions below, identify the terminology best suited:

This refers to the amount of the drug needed to produce a particular effect compared to another standard drug with similar receptor profile Choose...

This refers to the ability of the drug to bind to its appropriate receptor Choose...

This refers to how well the drug produces the expected response Choose...

Your answer is incorrect.

Potency of a drug with receptor binding action refers to the amount of the drug needed to produce a particular effect compared to another standard drug with similar receptor profile ('vigor').

Affinity refers to the ability of the drug to bind to its appropriate receptor ('affection').

Efficacy refers to how well the drug produces the expected response i.e. the maximum clinical response produced by a drug ('productivity').

Haloperidol is more potent than chlorpromazine - 5 mg of haloperidol can achieve the same effect as 100 mg of chlorpromazine. These drugs, however, produce equivalent maximal clinical response i.e. equally efficacious but not equipotent.

The correct answer is: This refers to the amount of the drug needed to produce a particular effect compared to another standard drug with similar receptor profile – Potency, This refers to the ability of the drug to bind to its appropriate receptor – Affinity, This refers to how well the drug produces the expected response – Efficacy

Question 73

Not answered

Marked out of 4.00

Flag question

MoK Neurosciences CXEMI025

Blood supply to the brain

Match the specific areas of the brain with their blood supply from the list above

1. Hypothalamus [▼] ✗

2. Basal ganglia [▼] ✗

3. Anterior limb of internal capsule (2 choices) [▼] ✗ [▼] ✗

Explanation

Anterior communicating and anterior medial arteries supply the hypothalamus.

The basal ganglia are supplied by the medial cerebral artery.

Anterior cerebral and anterior communicating arteries supply the anterior limb of the internal capsule.

The correct answer is:
Hypothalamus - Anterior communicating artery,

Basal ganglia - Medial cerebral artery,

Anterior limb of internal capsule (2 choices) - Anterior cerebral artery, Anterior communicating artery

Question 74

Not answered

Marked out of 4.00

Flag question

HiY Psychology EMI013

Memory types

Identify the type of memory which is being referred to, for each of the following situations.

A 45-year-old man was involved in a road traffic accident and was unconscious for a week. On examination, he has no memory for any events for a period of a week prior to the accident.	Choose... ▼
A 76-year-old man was found wandering and was brought to the A&E by the police. On mini-mental state examination, when you ask him to remember the three objects, he is unable to immediately repeat them	Choose... ▼
An 83-year-old woman has been diagnosed with Alzheimer's disease. She enjoys playing piano, which she used to do for a long time even before her illness began. Which type of memory remains intact?	Choose... ▼
When answering the question, 'What is the capital of France?' What is the type of memory Mr. P was using?	Choose... ▼

Your answer is incorrect.

Explanation: Retrograde amnesia involves the loss of memories that were created before the actual trauma or neurological damage.

Anterograde amnesia involves an inability to form new memories after the condition developed.

The immediate memory test in MMSE is thought to be tapping on to the domain of working memory.

Remembering a specific piano lesson is an example of explicit memory, while playing the piano as a result of this learning is an example of implicit procedural memory.

Semantic memory is the portion of long-term memory which is concerned with ideas, meanings, and concepts that are not related to personal experiences.

The correct answer is: A 45-year-old man was involved in a road traffic accident and was unconscious for a week. On examination, he has no memory for any events for a period of a week prior to the accident. – Retrograde memory, A 76-year-old man was found wandering and was brought to the A&E by the police. On mini-mental state examination, when you ask him to remember the three objects, he is unable to immediately repeat them – Working memory, An 83-year-old woman has been diagnosed with Alzheimer's disease. She enjoys playing piano, which she used to do for a long time even before her illness began. Which type of memory remains intact? – Procedural memory, When answering the question, 'What is the capital of France?' What is the type of memory Mr. P was using? – Semantic memory

Question 75

Not answered

Marked out of 4.00

Flag question

MoK Neurosciences EMI029

Neuroimaging tests in clinical practice

Identify the most likely neuroimaging findings for the following neuropsychiatric conditions

A 22-year-old gentleman with persecutory delusions and 3rd person auditory hallucinations. In addition to reduced frontotemporal and insular grey matter, what other feature can be seen in structural imaging?	Choose... ▼

A 69-year-old woman with recurrent episodes of low mood, anhedonia and poor sleep since the age of 60. What feature can be seen in structural imaging? Choose...

A 77-year-old man with history of short-term memory loss and poor concentration. In addition to ventricular enlargement, what other feature can be seen in structural imaging? Choose...

A 33-year-old woman with several waxing and waning neurological deficits involving ocular movements and sensorimotor system but intact cognitive abilities. Choose...

Your answer is incorrect.

Explanation:

Case 1: The diagnosis is schizophrenia. Neuroimaging findings in schizophrenia include ventricular enlargement and loss of grey matter - especially insula, anterior cingulate cortex, medial temporal lobe and dorsolateral pre-frontal cortex with some progressive loss of brain volume in the first few years of diagnosis. fMRI shows poor dorsolateral pre-frontal cortex activation in executive tasks, while decreased N-acetyl aspartate in prefrontal cortex is seen in spectroscopy.

Case 2: The diagnosis is depressive disorder (geriatric depression). Neuroimaging findings include periventricular and deep WM hyperintensities, subcortical thalamic and striatal hyperintensities, decreased frontal and basal ganglia volumes, and decreased metabolism in prefrontal cortex.

Case 3: The diagnosis is Alzheimer's dementia. Neuroimaging findings in Alzheimer's disease include ventricular enlargement, reduced medial temporal lobe volume - especially hippocampus and a reduction in parieto-temporal fMRI activation and SPECT blood flow.

Case 4: The diagnosis is Multiple Sclerosis. Neuroimaging findings in this disease include features of demyelination, especially white matter hypointensities affecting corpus callosum. If fluid-attenuated inversion recovery (FLAIR) MRI is used, periventricular high-signal intensity lesions can be observed.

The correct answer is: A 22-year-old gentleman with persecutory delusions and 3rd person auditory hallucinations. In addition to reduced frontotemporal and insular grey matter, what other feature can be seen in structural imaging?
– Ventricular enlargement, A 69-year-old woman with recurrent episodes of low mood, anhedonia and poor sleep since the age of 60. What feature can be seen in structural imaging? – White matter hyperintensities, A 77-year-old man with history of short-term memory loss and poor concentration. In addition to ventricular enlargement, what other feature can be seen in structural imaging? – Medial temporal atrophy, A 33-year-old woman with several waxing and waning neurological deficits involving ocular movements and sensorimotor system but intact cognitive abilities.
– White matter hypointensities

Question 76
Not answered
Marked out of 4.00
Flag question

MoK Neurosciences EMI030
Neuroanatomical lesions in CNS disorders
For each of the condition below choose the most specific option/s from given list

Korsakoff's psychosis Choose...

Gerstmann syndrome Choose...

Prosopagnosia Choose...

Alzheimer's disease Choose...

Your answer is incorrect.

Explanation:

About 80 per cent of alcoholic patients recovering from Wernicke's encephalopathy develop Korsakoff's amnesic syndrome. Diagnosis of Korsakoff's syndrome correlates best with the presence of lesions in the dorsomedial thalamus. But lesions in the mammillary bodies, the mammillothalamic tract and the anterior thalamus may be more relevant to memory dysfunction than lesions in the medial dorsal nucleus of the thalamus (Kopelamn et al., 2009).

Gerstmann syndrome is characterized by four primary symptoms: Dysgraphia/agraphia, Dyscalculia/acalculia, Finger agnosia and Left-right disorientation. This is often associated with brain lesions in the dominant (usually left) side of the angular and supramarginal gyri (parietal lobe).

Prosopagnosics cannot recognise familiar faces. Face processing is a bilateral function; more vital areas may be present on the right hemisphere. Acquired prosopagnosia is usually associated with bilateral or right-sided lesions of the occipital - temporal junction (fusiform gyrus).

In Alzheimer's disease, the main areas affected are the hippocampus and medial temporal lobe. Atrophic changes are mainly observed in these two areas in even in very early stages of Alzheimer's disease.

Barton, JJ., (2008). Prosopagnosia associated with a left occipitotemporal lesion. Excerpt retrieved from http://www.ncbi.nlm.nih.gov/pubmed/18374372
Kopeman et al., (2009). The Korsakoff Syndrome: Clinical Aspects, Psychology and Treatment. Excerpt retrieved from http://alcalc.oxfordjournals.org/content/44/2/148

The correct answer is: Korsakoff's psychosis
– Dorsomedial thalamus, Gerstmann syndrome – Supramarginal gyrus, Prosopagnosia – Occipitotemporal junction, Alzheimer's disease
– Hippocampus

Question 77
Not answered
Marked out of 2.00
Flag question

MoK Neurosciences EMI031
Neurohistology
Match the following neuroglial cells with their specific functions from the list provided:

Astrocytes Choose...

Ependymal cells Choose...

Your answer is incorrect.

Explanation: Astrocytes are the most numerous of all three types of glial cells. These are star-shaped cells that enable nutrition of neurons, breakdown of some neurotransmitters, and maintaining the blood-brain barrier.

Oligodendrocytes are seen in CNS (not in peripheral nerves, where Schwann cells replace them). They produce myelin sheaths that help in the saltatory conduction of an action potential (pole to pole jumping), which quicken the process of signal transmission.

The microglia are descendants of macrophages. They are scavenger cells that clear neuronal debris following cell death.

Ependymal cells are a particular type of glia that cover the ventricles and enable CSF circulation.

Schwann cells are seen in the peripheral nervous system where they produce myelin sheaths.

The correct answer is: Astrocytes
– Nutrition of neuronal cells, Ependymal cells – Covering the ventricles

Finish review